Q1.

An insurance company groups its vehicle insurance policies into two categories, car insurance and motorbike insurance.

The number of claims in a random sample of 80 policies was monitored and the results summarised in contingency **Table 1**.

Table 1

		Number of claims				
		0	**1**	**2**	**3 or more**	**Total**
Type of insurance policy	**Car**	9	10	11	5	35
	Motorbike	19	13	8	5	45
	Total	28	23	19	10	80

The insurance company decides to carry out a χ^2-test for association between number of claims and type of insurance policy using the information given in **Table 1**.

(a) The contingency table shown in **Table 2** gives some of the exact expected frequencies for this test.

Complete **Table 2** with the missing exact expected values.

Table 2

		Number of claims			
		0	**1**	**2**	**3 or more**
Type of insurance policy	**Car**		10.0625		4.375
	Motorbike			10.6875	

(2)

(b) Carry out the insurance company's test, using the 10% level of significance.

(8)
(Total 10 marks)

Q2.

A dairy industry researcher, Robyn, decided to investigate the milk yield, classified as low, medium or high, obtained from four different breeds of cow, A, B, C and D.

The milk yield of a sample of 105 cows was monitored and the results are summarised in contingency **Table 1**.

Table 1

		Yield			
		Low	**Medium**	**High**	**Total**
Breed	**A**	4	5	12	**21**
	B	10	6	4	**20**

	C	8	17	7	**32**
	D	5	20	7	**32**
	Total	**27**	**48**	**30**	**105**

The sample of cows may be regarded as random.

Robyn decides to carry out a χ^2-test for association between milk yield and breed using the information given in **Table 1**.

(a) Contingency **Table 2** gives some of the expected frequencies for this test.

Complete **Table 2** with the missing expected values.

Table 2

		Yield		
		Low	**Medium**	**High**
Breed	**A**			6
	B	5.14	9.14	5.71
	C			
	D	8.23	14.63	9.14

(2)

(b) (i) For Robyn's test, the test statistic $\sum \frac{(O-E)^2}{E} = 19.4$ correct to three significant figures.

Use this information to carry out Robyn's test, using the 1% level of significance.

(5)

(ii) By considering the observed frequencies given in **Table 1** with the expected frequencies in **Table 2**, interpret, in context, the association, if any, between milk yield and breed.

(2)
(Total 9 marks)

Q3.

A χ^2-test for association is carried out on frequency data given in a 5 × 3 contingency table using the 5% level of significance. All expected frequencies are greater than 5

State the number of degrees of freedom for this test.

Circle your answer.

6   8   14   15

(Total 1 mark)

Q4.

Students at a science department of a university are offered the opportunity to study an optional language module, either German or Mandarin, during their second year of study.

From a sample of 50 students who opted to study a language module, 31 were female. Of those who opted to study Mandarin, 8 were female and 12 were male.

Test, using the 5% level of significance, whether choice of language is independent of gender.

The sample of students may be regarded as random.

(Total 8 marks)

Q5.

A large estate agency would like all the properties that it handles to be sold within three months. A manager wants to know whether the type of property affects the time taken to sell it. The data for a random sample of properties sold are tabulated below.

	Type of property				
	Flat	**Terraced**	**Semi-detached**	**Detached**	**Total**
Sold within three months	4	34	28	18	84
Sold in more than three months	9	18	8	6	41
Total	13	52	36	24	125

(a) Conduct a χ^2-test, at the 10% level of significance, to determine whether there is an association between the type of property and the time taken to sell it. Explain why it is necessary to combine two columns before carrying out this test.

(10)

(b) The manager plans to spend extra money on advertising for one type of property in an attempt to increase the number sold within three months. Explain why the manager might choose:

(i) terraced properties;

(ii) flats.

(2)
(Total 12 marks)

Q6.

A town council wanted residents to apply for grants that were available for home insulation. In a trial, a random sample of 200 residents was encouraged, either in a letter or by a phone call, to apply for the grants. The outcomes are shown in the table.

	Applied for grant	Did not apply for grant	Total
Letter	30	130	160
Phone call	14	26	40
Total	44	156	200

(a) The council believed that a phone call was more effective than a letter in encouraging people to apply for a grant. Use a χ^2-test to investigate this belief at the 5% significance level.

(8)

(b) After the trial, all the residents in the town were encouraged, either in a letter or by a phone call, to apply for the grants. It was found that there was no association between the method of encouragement and the outcome. State, with a reason, whether a Type I error, a Type II error or neither occurred in carrying out the test in part (a).

(2)

(Total 10 marks)

Q7.

(a) **Table 1** contains the observed frequencies, a, b, c and d, relating to the two attributes, X and Y, required to perform a χ^2 test.

Table 1

	Y	**Not** Y	**Total**
X	a	b	m
Not X	c	d	n
Total	p	q	N

(i) Write down, in terms of m, n, p, q and N, expressions for the 4 expected frequencies corresponding to a, b, c and d.

(2)

(ii) Hence prove that the sum of the expected frequencies is N.

(3)

(b) Andy, a tennis player, wishes to investigate the possible effect of wind conditions on the results of his matches. The results of his matches for the 2011 season are represented in **Table 2**.

Table 2

	Windy	Not windy	Total
Won	15	18	33
Lost	12	5	17
Total	27	23	50

Conduct a χ^2 test, at the 10% level of significance, to investigate whether there is an association between Andy's results and wind conditions.

(8)

(Total 13 marks)

Q8.

Fiona, a lecturer in a school of engineering, believes that there is an association between the class of degree obtained by her students and the grades that they had achieved in A-level Mathematics.

In order to investigate her belief, she collected the relevant data on the performances of a random sample of 200 recent graduates who had achieved grades A or B in A-level Mathematics. These data are tabulated below.

		Class of degree				
		1	**2(i)**	**2(ii)**	**3**	**Total**
A-level grade	**A**	20	36	22	2	80
	B	9	55	48	8	120
	Total	29	91	70	10	200

(a) Conduct a χ^2 test, at the 1% level of significance, to determine whether Fiona's belief is justified.

(9)

(b) Make **two** comments on the degree performance of those students in this sample who achieved a grade B in A-level Mathematics.

(2)

(Total 11 marks)

Q9.

It is claimed that the way in which students voted at a particular general election was independent of their gender.

In order to investigate this claim, 480 male and 540 female students who voted at this general election were surveyed. These students may be regarded as a random sample.

The **percentages** of males and females who voted for the different parties are recorded in the table.

	Conservative	Labour	Liberal Democrat	Other parties
Male	32.5	30	25	12.5
Female	40	25	20	15

(a) Complete the contingency table below.

(2)

(b) Hence determine, at the 1% level of significance, whether the way in which students voted at this general election was independent of their gender.

	Conservative	Labour	Liberal Democrat	Other parties	Total
Male					480
Female					540
Total					1020

(9)

(Total 11 marks)

Q10.

Emily believed that the performances of 16-year-old students in their GCSEs are associated with the schools that they attend. To investigate her belief, Emily collected data on the GCSE results for 2010 from four schools in her area.

The table shows Emily's collected data, denoted by O_i, together with the corresponding expected frequencies, E_i, necessary for a χ^2 test.

	≥ 5 GCSEs		1 ≤ GCSEs < 5		No GCSEs	
	O_i	E_i	O_i	E_i	O_i	E_i
Jolliffe College for the Arts	187	193.15	93	90.62	30	26.23
Volpe Science Academy	175	184.43	97	86.52	24	25.05
Radok Music School	183	183.81	78	86.23	34	24.96
Bailey Language School	265	248.61	112	116.63	22	33.76

Emily used these values to correctly conduct a χ^2 test at the 1% level of significance.

(a) State the null hypothesis that Emily used.

(1)

(b) Find the value of the test statistic, χ^2, giving your answer to one decimal place.

(3)

(c) State, in context, the conclusion that Emily should reach based on the results of her χ^2 test.

(3)

(d) Make one comment on the GCSE performances of 16-year-old students attending Bailey Language School.

(1)

(e) Emily's friend, Joanna, used the same data to correctly conduct a χ^2 test using the 10% level of significance.

State, with justification, the conclusion that Joanna should reach.

(2)

(Total 10 marks)

Q11.

Julie, a driving instructor, believes that the first-time performances of her students in their driving tests are associated with their ages.

Julie's records of her students' first-time performances in their driving tests are shown in the table.

Age	Pass	Fail
17 – 18	28	20
19 – 30	2	14
31 – 39	12	33
40 – 60	6	5

(a) Use a χ^2 test at the 1% level of significance to investigate Julie's belief.

(9)

(b) Interpret your result in part (a) as it relates to the 17 – 18 age group.

(1)

(Total 10 marks)

Q12.

It is claimed that a new drug is effective in the prevention of sickness in holiday-makers. A sample of 100 holiday-makers was surveyed, with the following results.

	Sickness	No sickness	Total
Drug taken	24	56	80

No drug taken	11	9	20
Total	35	65	100

Assuming that the 100 holiday-makers are a random sample, use a χ^2 test, at the 5% level of significance, to investigate the claim.

(Total 8 marks)

Q13.

Fortune High School gave its students a wider choice of subjects to study. The table shows the number of students, of each gender, who chose to study each of the additional subjects during the school year 2007/08.

	Bulgarian	**Climate Change**	**Finance**	**Polish**
Male	7	31	25	40
Female	2	24	22	19

Assuming that these data form a random sample, use a χ^2 test, at the 10% level of significance, to test whether the choice of these subjects is independent of gender.

(Total 11 marks)

Q14.

A sample survey, conducted to determine the attitudes of residents to a proposed reorganisation of local schools, gave the following results.

		Against reorganisation	**Not against reorganisation**
Age of resident	**16–17**	9	2
	18–21	17	10
	22–49	115	90
	50–65	41	34
	Over 65	3	4

Use a χ^2 test, at the 5% level of significance, to determine whether there is an association between the ages of residents and their attitudes to the proposed reorganisation of local schools.

(Total 12 marks)

Q15.

A survey is carried out in an attempt to determine whether the salary achieved by the age of 30 is associated with having had a university education.

The results of this survey are given in the table.

	Salary < £30 000	Salary ≥ £30 000	Total
University education	52	78	130
No university education	63	57	120
Total	115	135	250

(a) Use a χ^2 test, at the 10% level of significance, to determine whether the salary achieved by the age of 30 is associated with having had a university education.

(9)

(b) What do you understand by a Type I error in this context?

(2)

(Total 11 marks)

Q16.

It is thought that the incidence of asthma in children is associated with the volume of traffic in the area where they live.

Two surveys of children were conducted: one in an area where the volume of traffic was heavy and the other in an area where the volume of traffic was light.

For each area, the table shows the number of children in the survey who had asthma and the number who did not have asthma.

	Asthma	No asthma	Total
Heavy traffic	52	58	110
Light traffic	28	62	90
Total	80	120	200

(a) Use a χ^2 test, at the 5% level of significance, to determine whether the incidence of asthma in children is associated with the volume of traffic in the area where they live.

(8)

(b) Comment on the number of children in the survey who had asthma, given that they lived in an area where the volume of traffic was heavy.

(1)

(Total 9 marks)

Q17.

A statistics unit is required to determine whether or not there is an association between students' performances in mathematics at Key Stage 3 and at GCE.

A survey of the results of 500 students showed the following information:

		GCE Grade				
		A	B	C	Below C	Total
Key Stage 3 Level	8	60	55	47	43	205
	7	55	32	31	26	144
	6	40	38	35	38	151
	Total	155	125	113	107	500

(a) Use a χ^2 test at the 10% level of significance to determine whether there is an association between students' performances in mathematics at Key Stage 3 and at GCE.

(9)

(b) Comment on the number of students who gained a grade A at GCE having gained a level 7 at Key Stage 3.

(1)
(Total 10 marks)

Q18.

Two groups of patients, suffering from the same medical condition, took part in a clinical trial of a new drug. One of the groups was given the drug whilst the other group was given a placebo, a drug that has no physical effect on their medical condition.

The table shows the number of patients in each group and whether or not their condition improved.

	Placebo	Drug
Condition improved	20	46
Condition did not improve	55	29

Conduct a χ^2 test, at the 5% level of significance, to determine whether the condition of the patients at the conclusion of the trial is associated with the treatment that they were given.

(Total 10 marks)

Q19.

It is claimed that the area within which a school is situated affects the age profile of the staff employed at that school. In order to investigate this claim, the age profiles of staff employed at two schools with similar academic achievements are compared.

Academia High School, situated in a rural community, employs 120 staff whilst Best Manor Grammar School, situated in an inner-city community, employs 80 staff.

The **percentage** of staff within each age group, for each school, is given in the table.

Age	Academia High School	Best Manor Grammar School
22-34	17.5	40.0
35-39	60.0	45.0
40-59	22.5	15.0

(a) (i) Form the data into a contingency table suitable for analysis using a χ^2 distribution.

(2)

(ii) Use a χ^2 test, at the 1% level of significance, to determine whether there is an association between the age profile of the staff employed and the area within which the school is situated.

(9)

(b) Interpret your result in part (a)(ii) as it relates to the 22-34 age group.

(2)

(Total 13 marks)

Q20.

A hockey team's coach believes that the results of matches are affected by whether the team plays at home or away.

The results of 50 randomly selected matches played by the team are given in the table below.

	Home	Away	Total
Win	15	5	20
Lose	6	12	18
Draw	6	6	12
Total	27	23	50

Use a χ^2 test, at the 5% level of significance, to determine whether the coach's belief is justified.

(Total 10 marks)

Q21.

A survey of 420 single people was conducted in order to find out whether there is evidence of an association between their gender and the type of holiday that they prefer.

The results of this survey are as follows.

	Beach	Adventure

	holiday	holiday
Male	118	98
Female	90	114

Investigate, at the 5% level of significance, the claim that there is an association between the gender of a single person and the type of holiday that they prefer.

(Total 10 marks)

Q22.

Baljeet's parents want to choose a school for her to attend in order to pursue her sixth form studies. They have a choice of sending her to one of two schools, X or Y. To help them make an informed choice, they decide to look at the numbers of A, B, C and D grades achieved last year by each school. These are tabulated below.

	Examination Grades				
	A	**B**	**C**	**D**	**Total**
School X	52	34	16	18	120
School Y	114	58	62	46	280
Total	166	92	78	64	400

Stating the null hypothesis, carry out a χ^2 test at the 5% level of significance to determine whether there is an association between the schools and the numbers of A, B, C and D grades achieved.

(Total 9 marks)

Q23.

The results of a recent police survey of traffic travelling on motorways produced information about the genders of drivers and the speeds, S miles per hour, of their vehicles, as tabulated below.

	Speed of vehicle		
	$S \le 70$	$70 < S \le 90$	$S > 90$
Male	17	40	70
Female	30	25	18

Stating null and alternative hypotheses, investigate, at the 1% level of significance, the claim that there is no association between the gender of the driver and the speed of the vehicle.

(Total 11 marks)

Q24.

For her 17th birthday present, Susan wishes to have a course of driving lessons. In an attempt to select the best driving school in the area, her parents compare recent test results of two schools, A and B.

These test results are tabulated below.

	School A	School B
Pass	120	100
Fail	24	36

Stating the null hypothesis, use a χ^2 test at the 5% level of significance to determine whether there is an association between test results and driving school.

(Total 9 marks)

Q25.

The prisoners in a high security jail are placed into three categories: Short stay, Medium stay and Long stay. The frequency with which the prisoners receive mail is monitored and the results recorded in the table below.

	Length of stay in prison			
	Short	**Medium**	**Long**	
Regular mail	40	24	8	72
Occasional mail	12	18	15	45
No mail	13	27	22	62
	65	69	45	179

Stating your null and alternative hypotheses, investigate, at the 1% level of significance, the claim that there is no association between length of stay in prison and the frequency with which prisoners receive mail.

(Total 10 marks)

Q26.

The time taken to complete a tennis match varies. Boris is a professional tennis player who believes that, because of his excellent physical condition, the longer a match lasts the better his chance of winning. An analysis of a sample of matches in which Boris had taken part showed that:

of 43 matches which lasted less than two hours, Boris had won 28;
of 52 matches which lasted two hours or longer, Boris had won 22.

(a) Form the data into a 2 x 2 contingency table suitable for analysis using a χ^2 distribution.

(3)

(b) Analyse the table that you have formed in part (a) using the 5% significance level.

(9)

(c) Interpret your results in parts (a) and (b) in the context of the question.

(2)
(Total 14 marks)

Q27.

A political party held a conference, attended by delegates from local branches, to choose a new leader. The voting at the conference was carried out in public. In a confidential survey, randomly selected delegates were asked whether they would have voted differently if voting had been by secret ballot. The answers, classified by the ages of the delegates, are summarised in the following table.

	Definitely	**Possibly**	**Definitely not**
Under 65 years	**4**	**12**	6
65 years and over	**10**	**28**	54

(a) Use a χ^2 distribution and the 5% significance level to examine whether the answer to the question is associated with the age of the delegate.

(10)

(b) Interpret your result in part (a) in the context of the question.

(1)

(c) Make a further comment about the ages of the delegates.

(1)
(Total 12 marks)

Q28.

Chandra, an estate agent, sells properties in a large conurbation. She classifies the areas of the conurbation as:

City Centre;

Inner City – close to but outside the City Centre;
Suburban – outside the Inner City.

The following table shows the number of properties she has for sale in each of these areas together with information on the Advertised Price.

		Area		
		City Centre	**Inner City**	**Suburban**
Advertised Price	**< £120 000**	2	58	20
	≤ £120 000	34	12	44

(a) Use a χ^2 distribution and the 1% significance level to analyse this contingency table.

(9)

(b) Interpret your result in the context of this question.

(2)
(Total 11 marks)

Q29.

The board of an insurance company was concerned about the poor performance of its motor insurance business. Belinda was asked to collect and analyse data on the claims record of the 4902 customers who held motor policies in 2003.

(a) The following contingency table summarises the number of claims made in 2003 classified by length of time the policy had been held.

		Number of Claims in 2003			
		0	**1**	**2**	**3 or more**
Policy held for	**less than 2 years**	1478	212	21	6
	2 or more years	2830	334	20	1

Use a χ^2 distribution and the 1% significance level to investigate whether the number of claims made in 2003 is associated with the length of time for which the policy had been held.

(11)

(b) Belinda also constructed the following tables concerning the 4902 customers.

Table A

		% of Total Number of Policy Holders who in 2003	
		Made a Claim	**Did not make a Claim**
Gender	**Male**	8.4	57.6
	Female	3.7	30.3

Table B

		Total Number of Policy Holders in 2003	**Number of Policy Holders who made a claim in 2003**
Age (years)	**< 25**	956	280
	25 or more	3946	314

Table C

Number of Policy Holders who in 2003	
Did not make a Claim	**Made a claim for more than £2000**

Age	**< 25**	676	145
(years)	**25 or more**	3632	192

For **each** table:

(i) state why, in its present form, it cannot be analysed as a contingency table using a χ^2 distribution;

(3)

(ii) construct a new table which can be analysed as a contingency table using a χ^2 distribution. **Do not analyse any of the tables you construct**.

(6)
(Total 20 marks)

Q30.

Rhamesh runs a café which opens at noon and admits no further customers after 9 pm. He does not sell alcoholic drinks, but customers can bring their own beer or wine with them, to drink with their meal, if they wish.

To help him decide whether it would be worthwhile selling his own alcoholic drinks, Rhamesh asks Anna, a student, to record how many customers bring their own alcoholic drinks. Anna suspects this may be related to the time of day, and following her observations she produces the table below.

The table shows the number of customers, with and without alcoholic drinks, who entered the café at different times on a particular day.

	Time of Day		
	Noon to 3 pm	**3 pm to 6 pm**	**6 pm to 9 pm**
Alcoholic drink	12	6	54
No alcoholic drink	62	35	67

(a) Use the 1% significance level to investigate whether taking alcohol with a meal is independent of time of day.

(9)

(b) (i) Interpret your result in part (a) as it relates to customers entering the cafe between 6 pm and 9 pm.

(ii) Make a further statement about customers entering the caf6 between 6 pm and 9 pm.

(3)

(c) How might your conclusions be affected, if at all, by the fact that the table shows the customers who entered the cafe on only one particular day?

(2)
(Total 14 marks)

Q31.

A well known picture of the Beatles shows them on a pedestrian crossing outside the Abbey Road recording studios. Substantial numbers of Beatles fans visit Abbey Road and use the pedestrian crossing.

It was claimed that these fans were causing delays to rush hour traffic. As part of an investigation, people using the crossing were asked whether they were using it in the course of their normal daily lives or because of the Beatles photograph. The answers and the times of day are summarised in the contingency table below.

	Time of day	
	Rush hour	**Out of rush hour**
Normal daily lives	34	45
Because of Beatles	4	28

(a) Using the 5% significance level, examine whether the reason for using the crossing is associated with the time of day.

(10)

(b) Interpret your result in the context of this question.

(1)
(Total 11 marks)

Q32.

Students following a media studies course at a particular university must study a foreign language. They are given a choice of Japanese, Russian or Swedish. The following table summarises the choices by gender of a random sample of students entering the course.

	Japanese	Russian	Swedish
Female	16	12	22
Male	12	14	54

(a) Use a χ^2 distribution and the 5% significance level to investigate whether choice of language is associated with gender.

(10)

(b) (i) Interpret your result in part (a) as it relates to the choice of Swedish.

(ii) Make a further statement about the choice of Swedish.

(3)
(Total 13 marks)

Q33.

A railway company wishes to monitor passenger opinion. During a pilot study a randomly selected sample of rail passengers was surveyed. As part of the survey they were asked whether they carried mobile phones for use on their journey and whether they were irritated by other passengers using mobile phones on the journey.

The results are summarised in the following table.

	Irritated	Not irritated
Mobile phone carried	12	14
Mobile phone not carried	26	6

(a) (i) Use a χ^2 test, at the 5% significance level, to analyse the contingency table above.

(10)

(ii) Interpret your conclusion in the context of the question.

(1)

(b) As a result of the pilot survey the questionnaire was modified and a further random sample of rail passengers was asked to complete it. The answers to the questions relating to mobile phones are summarised in the following table.

	Not irritated	Slightly irritated	Irritated	Extremely irritated
Mobile phone carried	6	46	18	3
Mobile phone not carried	3	7	14	17

The expected values for the analysis of this contingency table are given below.

	Not irritated	Slightly irritated	Irritated	Extremely irritated
Mobile phone carried	5.76	33.94	20.49	12.81
Mobile phone not carried	3.24	19.06	11.51	7.19

Jason is asked to analyse the data and suggests the “Not irritated” and “Extremely irritated” columns be combined together before carrying out the analysis.

Eric suggests that it would be better to combine the “Not irritated” column with the “Slightly irritated” column and the “Extremely irritated” column with the “Irritated” column.

Anthony suggests that the “Not irritated” column should be combined with the “Slightly irritated” column and the resulting 2×3 table analysed.

(i) Give one criticism of Jason’s suggestion.

(1)

(ii) Explain why Anthony’s suggestion is preferable to Eric’s.

(2)

(iii) Carry out Anthony’s suggestion, using the 1% significance level, and interpret your conclusion in context.

(8)

(Total 22 marks)